



Estimating EKS Costs

Estimate an architecture's monthly cost before building it: list every billable resource, price the fixed ones exactly, and bound the usage-driven ones with realistic assumptions.

IN THIS LECTURE

01 The meter model

What a cloud bill actually is.

02 Fixed vs. usage-driven

The one distinction that matters.

03 Categories and traps

What bills are made of, where surprises hide.

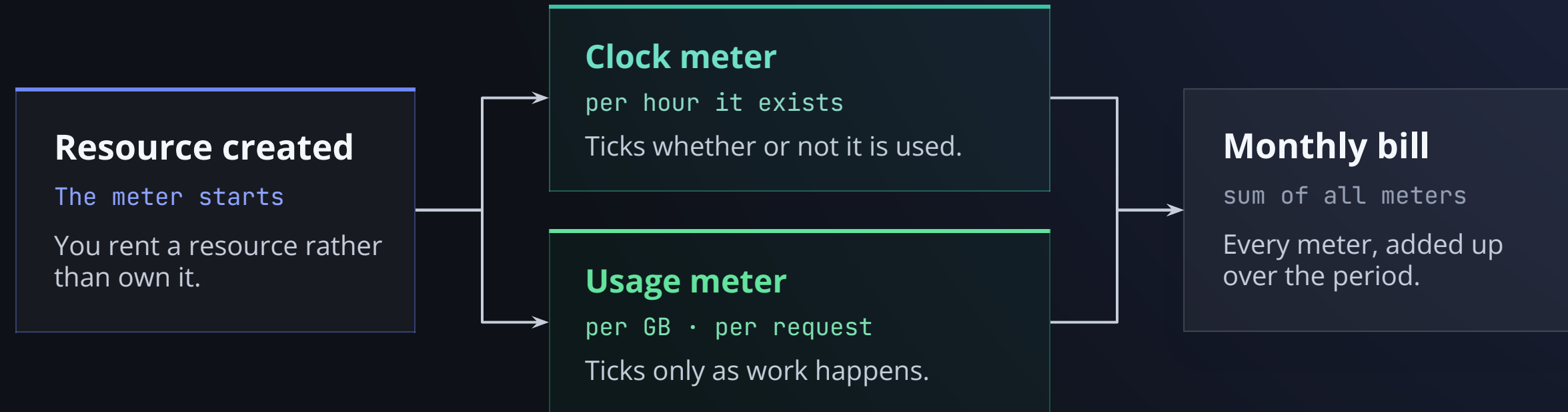
04 A method and the tools

Five steps, plus calculator and cost explorer.



Every resource runs a meter

Create a resource and a meter starts. Your monthly bill is nothing more than the sum of every meter over the billing period.



A forgotten resource **still runs its clock**. The charge stops when you delete the resource, not when you stop using it.

THE KEY DISTINCTION

Fixed cost vs usage-driven cost

Sort every meter into one of two buckets, and an intimidating bill becomes a floor you know plus a variable you can bound.

Fixed cost

the floor

- Clock-based: accrues just because the resource exists.
- Managed-service fees, gateways, idle load balancers, allocated addresses.
- Computed exactly, with no guessing.
- What you pay at 3 a.m. with zero traffic.

`hourly rate × ~730 h = exact floor`

Usage-driven cost

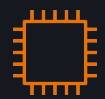
the variable

- Scales with what the workload actually does.
- GB stored, GB moved, requests served.
- Cannot be computed exactly in advance.
- Estimate with a realistic assumption, then put a range around it.

`assumption × traffic = bounded range`

Six categories in every bill

Every resource on the bill falls into one of these six, so classifying them is how you read an unfamiliar architecture's cost.



Compute

Rented servers, billed on the clock while they run.



Networking

Data moved out to the internet and across zones.



Encryption keys

A small monthly fee per key, plus per-request charges.



Storage

Disks and buckets, billed per GB per month.



Managed-service fees

Flat per-hour or per-cluster charges for a service just existing.



Addresses, logs, monitoring

The small line items that quietly add up.

WHERE COSTS HIDE

The five costs people miss

None of these are huge alone. Together they turn a confident estimate into a wrong one.

! **Egress to the internet:** billed per GB, and invisible on the architecture diagram.

! **Cross-zone traffic:** a healthy, spread-out cluster pays per GB in each direction.

! **Gateways bill twice:** once per hour to exist, again per GB processed.

! **Idle but running:** a gateway or load balancer you stopped using still bills at full clock rate.

! **Public addresses are not free:** every held public address now carries a small hourly charge.

! Name these **explicitly on every estimate**, because they are where a confident number quietly goes wrong.

THE METHOD

Estimate in five steps

Steps one to three you compute exactly. Steps four and five are judgment calls you refine as real usage data arrives.



■ Steps 1 to 3: computed exactly ■ Steps 4 to 5: estimated, then bounded

THE TOOLS

Calculator before, Cost Explorer after

Estimate first, compare to reality later, and let the gap sharpen your next estimate.



Pricing calculator

before you build

- ✓ A web tool: add each service, set quantities and region.
- ✓ Returns a line-by-line monthly estimate you can save and share.
- ✓ Forces you to enumerate and classify as you go.
- ✓ Holds current per-region rates for you.



Cost Explorer

after you build

- ✓ Shows what you have actually spent, by service.
- ✓ Breaks down by project or environment when resources are tagged.
- ✓ The reality check against your estimate.
- ✓ The gap tells you where your assumptions were wrong.

estimate



compare



Feed each gap back into your next estimate, and the numbers get sharper over time.